GI Pathology 1.3 Acute and Chronic Gastritis & Peptic Ulcer Disease



1. **Damaged peptic concrete** : acute gastropathy (e.g. damaged gastric mucosa due to deficient mucus lining or direct toxic effect)
2. **Fire extinguisher** : NSAIDs can cause acute gastropathy (direct toxic effect; AND inhibition of COX-1 → decreased prostaglandin synthesis → decreased mucus production)
3. **Smoking and drinking** : both alcohol and tobacco have direct toxic effects on the gastric epithelium, which can lead to acute gastropathy
4. **Flaming peptic concrete** : acute gastropathy can lead to acute gastritis (vascular congestion, edema, and neutrophilic infiltration on histology)
5. **Leaking shallow hole** : severe damage may lead to an EROSION which extends through the gastric epithelium into the lamina propria (may cause acute hemorrhagic erosive gastropathy)
6. **Leaking deep hole through rebar layer** : gastric ulcers are erosions that have extended through the muscularis mucosa into the submucosa (risk for bleeding)
7. **Burning curly drill bit** : Curling ulcers can develop with severe burns
8. **Dumping bucket on burn** : Curling ulcers develop due to hypovolemia and decreased gastric mucosal perfusion
9. **Crushing head** : Cushing ulcers can develop with high intracranial pressure
10. **Vegas sign** : increased intracranial pressure causes massive vagal output → increased gastric acid production → Cushing ulcers
11. **Stomach on fire** : acute gastritis can cause vague epigastric pain (dyspepsia)
12. **Nauseated** : acute gastritis can cause nausea, vomiting, anorexia
13. **Chronic grandfather clock** : chronic gastritis (e.g. H. pylori infection, autoimmune gastritis)
14. **Helicopter** : H. pylori infection is the most common cause of chronic gastritis
15. **Helicopter sinking through mucus** : H. pylori invades the stomach's mucus layer to reside on the gastric epithelium
16. **Pink commas** : H. pylori is a comma-shaped gram-negative bacilli
17. **Helicopter adjacent to distal stomach** : H. pylori usually infects the antrum first
18. **Fire under helicopter** : initial infection with H. pylori can cause acute gastritis
19. **Spotlight pointing across the stomach** : H. pylori infection spreads from the antrum to the body of the stomach
20. **Chronic shack roof on fire** : chronic H. pylori infection causes inflammation of gastric mucosa
21. **First responders** : chronic inflammation due to H. pylori infection displays a prominent neutrophil infiltrate
22. **Thinning fabric** : chronic inflammation from H. pylori leads to chronic atrophic gastritis (mucosal atrophy and loss of acid- and hormone-secreting glands → decreased gastrin and acid secretion)
23. **White mucus knots in columnar fence** : chronic inflammation from H. pylori leads to intestinal metaplasia (normal gastric columnar epithelium develops goblet cell hyperplasia)
24. **Intestinal metal pipes** : chronic inflammation from H. pylori leads to intestinal metaplasia
25. **Cancer crab; intestinal metaplasia (due to chronic inflammation from H. pylori) can progress to dysplasia and cancer** :
26. **Pile of lymphoid tissues** : chronic H. pylori infection can lead to development of Mucosa-Associated Lymphoid Tissue (MALT) (chronic antigenic presence of H. pylori causes accumulation of lymphocytes, which organize into lymphoid tissue with germinal centers)
27. **Blue tissues** : on histology, MALT appear blue due to the presence of lymphocytes
28. **Crab on helicopter** : MALT (caused by H. pylori infection) can transform into gastric lymphoma
29. **Biopsy grabber** : H. pylori infection can be diagnosed with gastric biopsy (antrum and body)
30. **Ammonia spray bottle** : urease testing of gastric biopsy specimen increases sensitivity of testing for H. pylori
31. **Gas from ammonia bottle** : H. pylori can be diagnosed with a urea breath test (patient drinks radiolabeled urea → urease in H. pylori metabolizes urea into radiolabeled CO_2 → patient breathes CO_2 into a detector)

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32. **Antibody forklift** : in autoimmune metaplastic atrophic gastritis, autoantibodies cause destruction of parietal cells and intrinsic factor
33. **Targeting parietal protesters** : parietal cells (located in body of stomach) are targeted in autoimmune metaplastic atrophic gastritis
34. **Dropping lemons** : in autoimmune metaplastic atrophic gastritis, destruction of parietal cells causes decreased secretion of gastric acid
35. **Oversized "GAS" can** : in autoimmune metaplastic atrophic gastritis, gastrin levels are increased (destruction of parietal cells → decreased HCl secretion → hyperplasia of gastrin-producing G cells in antrum → hypergastrinemia)
36. **"NO INTRINSIC right"** : in autoimmune metaplastic atrophic gastritis, intrinsic factor is decreased (normally binds to vitamin B12 to facilitate its absorption)
37. **Mushroom cloud "blast"** : in autoimmune metaplastic atrophic gastritis, decreased B12 absorption can cause pernicious anemia (a megaloblastic anemia [large, fragile erythrocytes])
38. **Jackhammer eroding through rebar layer**: peptic ulcer disease (PUD) (ulcers penetrate through the muscularis mucosa into the submucosa)
39. **P.U.D. helicopter** : PUD usually develops over a background of chronic gastritis (e.g. H. pylori infection)
40. **Lesser curvature hole** : gastric ulcer (most commonly occur in the lesser curvature of the stomach)
41. **Distal duodenal hole** : duodenal ulcer (more common than gastric ulcer)
42. **Knocked-over STOP sign** : H. pylori-mediated chronic gastritis leads to the destruction of antral delta cells → decreased somatostatin
43. **GAS tanks** : in chronic gastritis (e.g. H. pylori infection), loss of somatostatin leads to uninhibited gastrin production
44. **Activated protesters throwing lemons at gas guy** : increased gastrin stimulates acid secretion by parietal cells
45. **Lemon juice in duodenal hole** : in chronic gastritis (e.g. H. pylori infection), increased acid delivery to the duodenum leads to duodenal erosion and ulcers
46. **Fire extinguisher** : NSAID use is a risk factor for peptic ulcer disease in the STOMACH (SYNERGISTIC with H. Pylori)

47. **Smoker** : smoking is a risk factor for peptic ulcer disease (both gastric and duodenal)
48. **Eating lunch with active jackhammer** : in GASTRIC peptic ulcer disease, epigastric pain is WORSE with meals (due to increased gastrin-mediated acid secretion)
49. **Eating lunch with inactive jackhammer** : in DUODENAL peptic ulcer disease, epigastric pain is IMPROVED with meals (due to decreased passage of acid into the duodenum after food intake, and release of bicarbonate from the pancreas)
50. **Hole leaking blood** : peptic ulcer disease can cause upper GI bleeding (e.g. melena or hematemesis if severe)
51. **Black tar** : peptic ulcer disease can cause melena ("black tarry stools" due to oxidation of iron from hemoglobin breakdown), a complication of UPPER GI bleeds (proximal to suspensory ligament of the duodenum)
52. **Big gastric blood pipe** : gastric ulcers may bleed heavily (due to erosion into left gastric artery)
53. **"Path Obstructed"** : gastric outlet obstruction is a complication of peptic ulcer disease (due to inflammation causing swelling of pylorus)(presents with nonbilious emesis)
54. **Big duodenal air pipe** : duodenal ulcers can lead to perforation (most common in anterior duodenum)
55. **Small gastric air pipe** : gastric ulcers can lead to perforation (less common than in duodenal ulcers)
56. **Suspender strap popping** : ulcer perforation causes shoulder pain (acute peritonitis → irritation of diaphragm and phrenic nerve → pain in C3-C5 dermatomes)
57. **Black belt** : ulcer perforation causes radiolucency (i.e. darkness) between right hemidiaphragm and liver on upright abdominal X-ray (due to free air in abdominal cavity)
58. **Grabbing crab** : peptic ulcers and surrounding areas are biopsied to rule-out malignancy